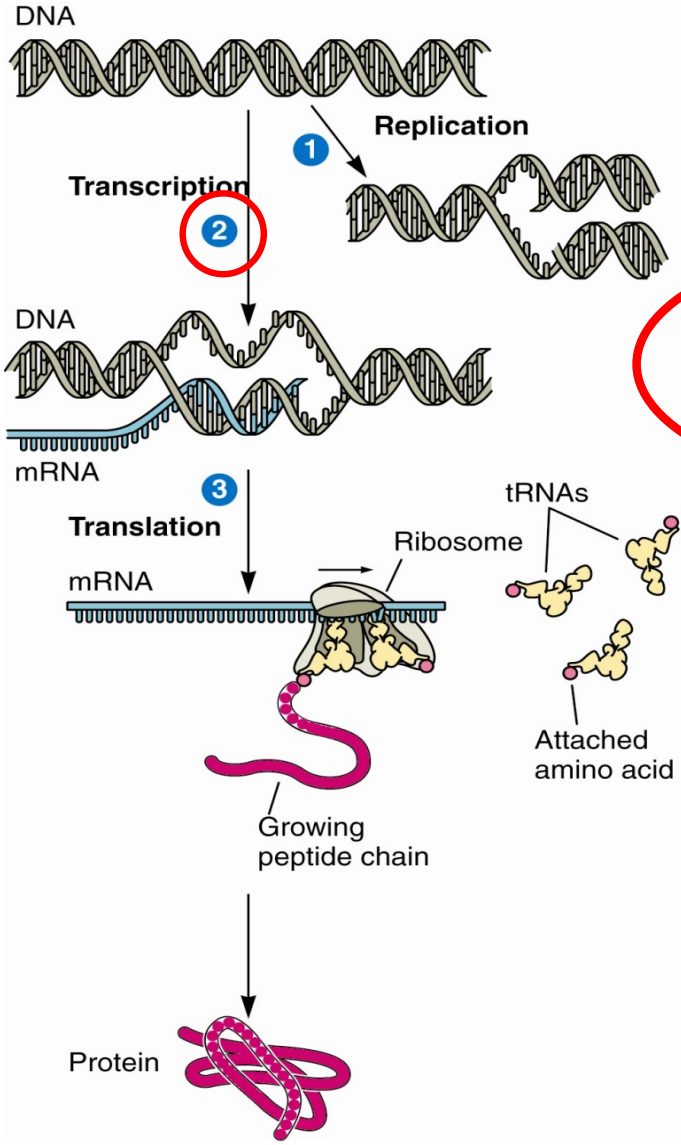
Transcription

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The “central dogma” of molecular biology



1 Replication
DNA replication yields two DNA molecules identical to the original one, ensuring transmission of genetic information to daughter cells with exceptional fidelity.

2 Transcription
The sequence of bases in DNA is recorded as a sequence of complementary bases in a single-stranded mRNA molecule.

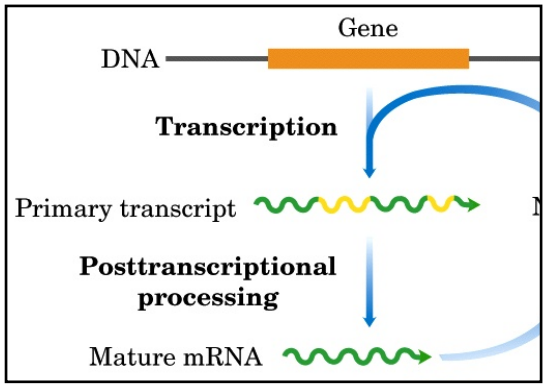
3 Translation
Three-base codons on the mRNA corresponding to specific amino acids direct the sequence of building a protein. These codons are recognized by tRNAs (transfer RNAs) carrying the appropriate amino acids. Ribosomes are the “machinery” for protein synthesis.

TRANSCRIPTION

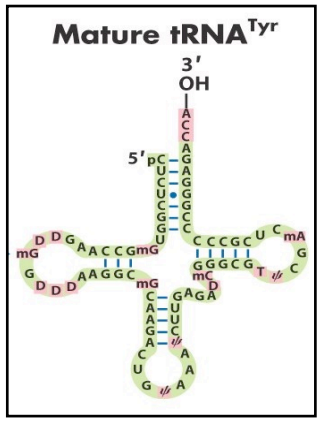
The synthesis of RNA molecules using the information contained in the DNA sequence.

There are four fundamental classes of RNA:

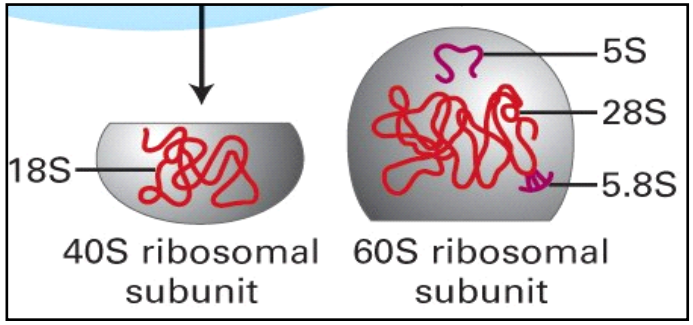
1. Messenger RNA (mRNA).



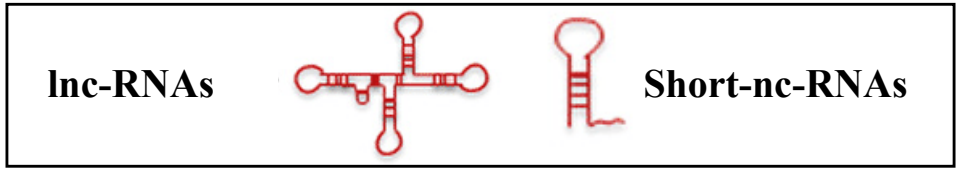
2. Transfer RNA (tRNA) .



3. Ribosomal RNA (rRNA).

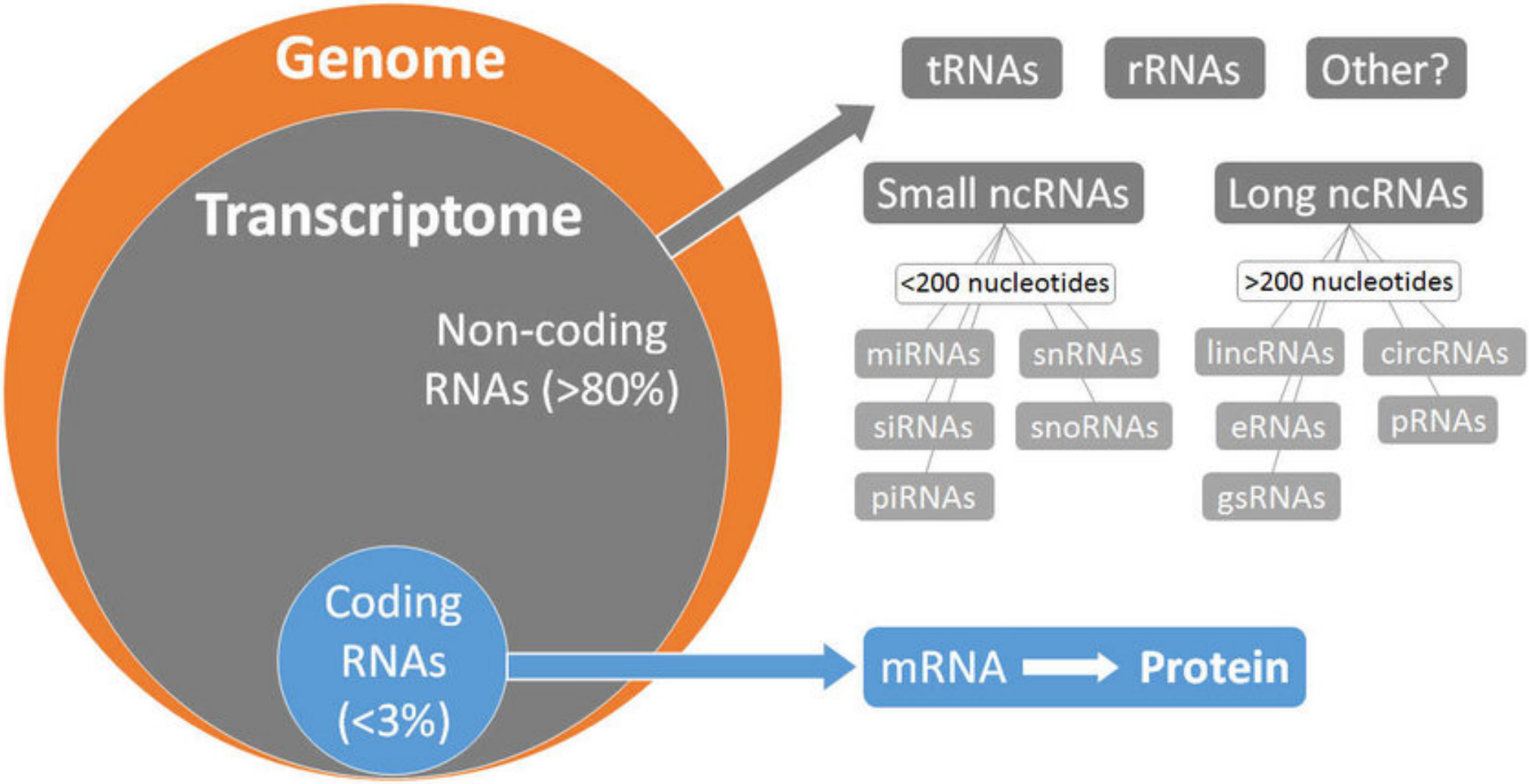


4. Non-coding RNAs (ncRNAs):



The complement of RNA molecules produced in a cell under specific conditions is termed TRANSCRIPTOME

Coding and non-coding RNAs in the human genome

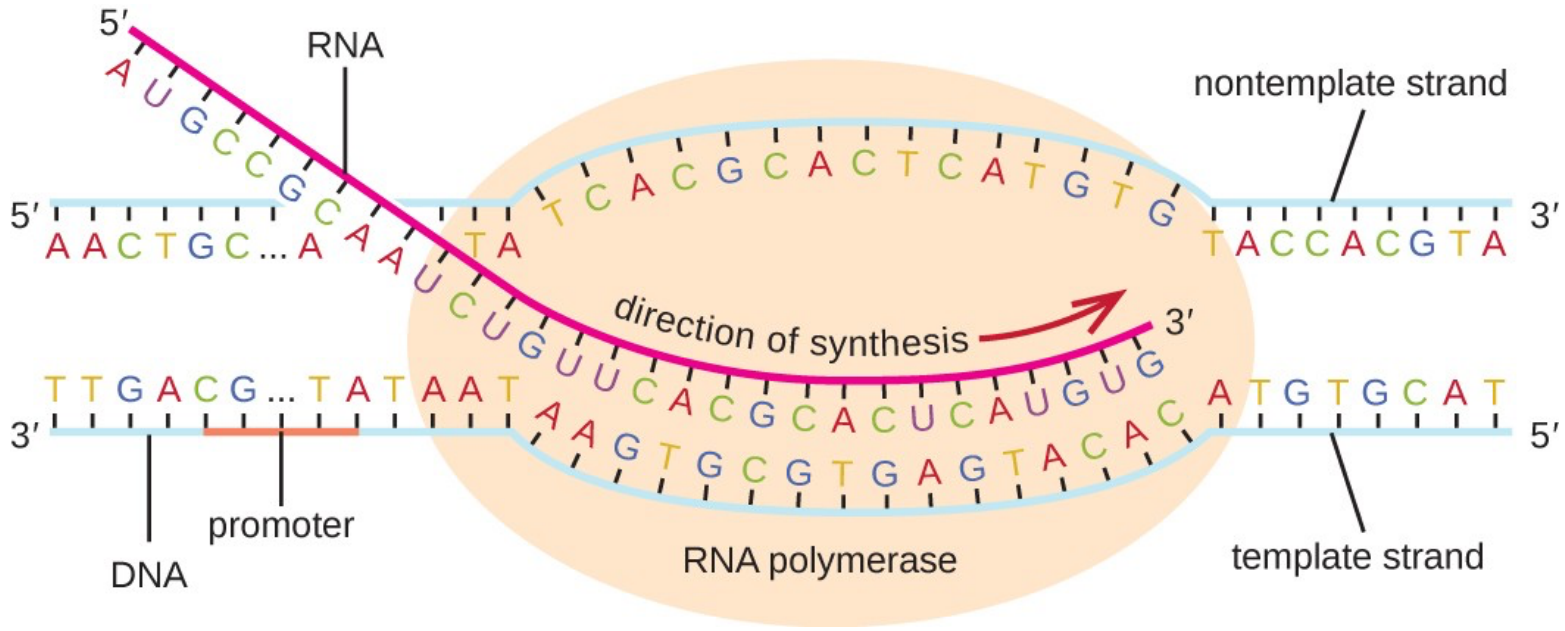


RNA synthesis is a polymerization reaction catalyzed by enzymes known as "RNA polymerases"

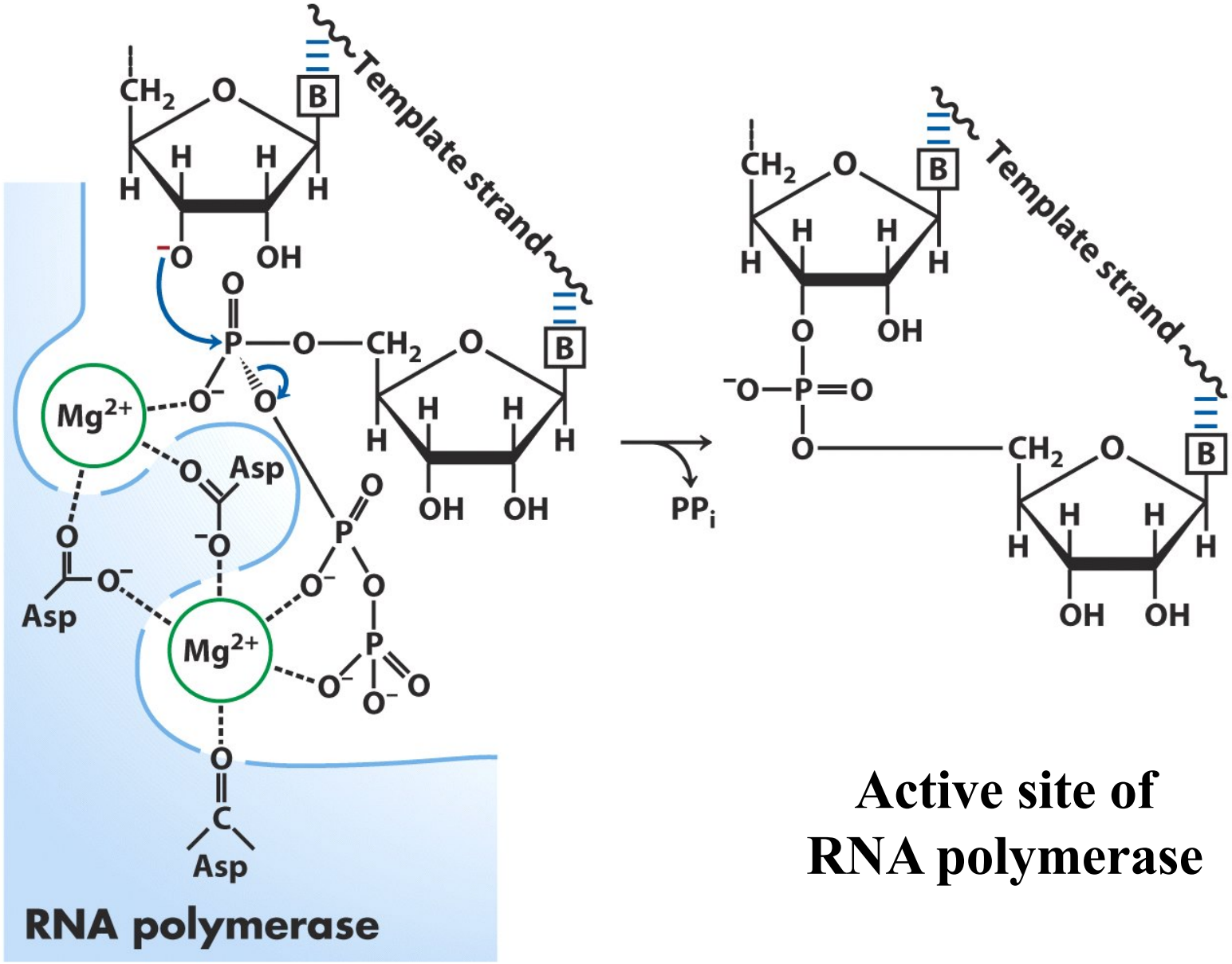
DNA template + n(NTP) → $pppN(pN)_n$+(n-1)PPi+ DNA template

NTPs = ATP, GTP, CTP, UTP.

- RNA synthesis involves the incorporation of nucleotides and the release of PPi.
- RNA synthesis is guided by a DNA template.
- RNA synthesis occurs in 5' to 3' direction.



Catalytic mechanism of RNA polymerases



Active site of RNA polymerase

Prokaryotic RNA polymerase: functions

1. Identifies the DNA sequences that mark the start of transcription.

2. Unwinds the DNA double helix to expose the template.

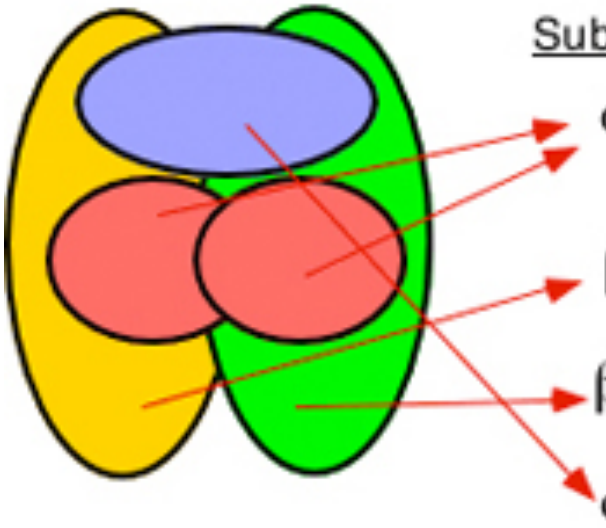
3. Selects the correct nucleotide and catalyzes the formation of phosphodiester bond: 5’ to 3’ processive polymerase activity (does not have proofreading activity).

4. Detects termination signals that mark the end of transcription.

5. Interacts with activating and repressing proteins that modulate transcription’s speed.

Prokaryotic RNA polymerase

Catalyzes the synthesis of mRNA, rRNA, tRNA



Subunit	Size	#/Molecule	Function
α	36.5 kD	2	chain initiation and interaction with regulatory proteins
β	151 kD	1	chain initiation and elongation
β'	155 kD	1	DNA binding
σ	70 kD	1	promoter recognition

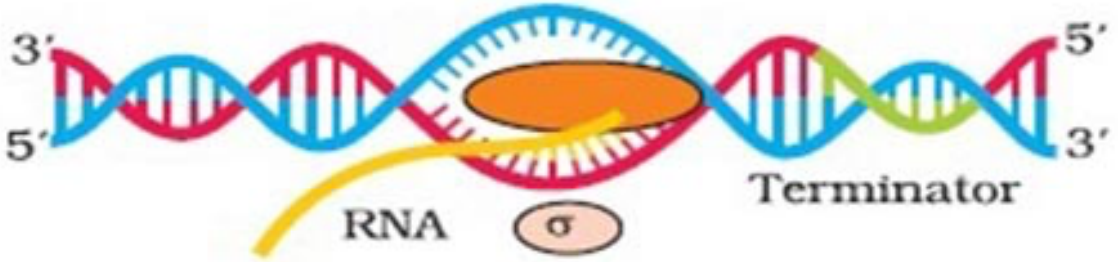
Mol. weight ≈ 390 kD

Transcription involves three sequential steps:

Initiation



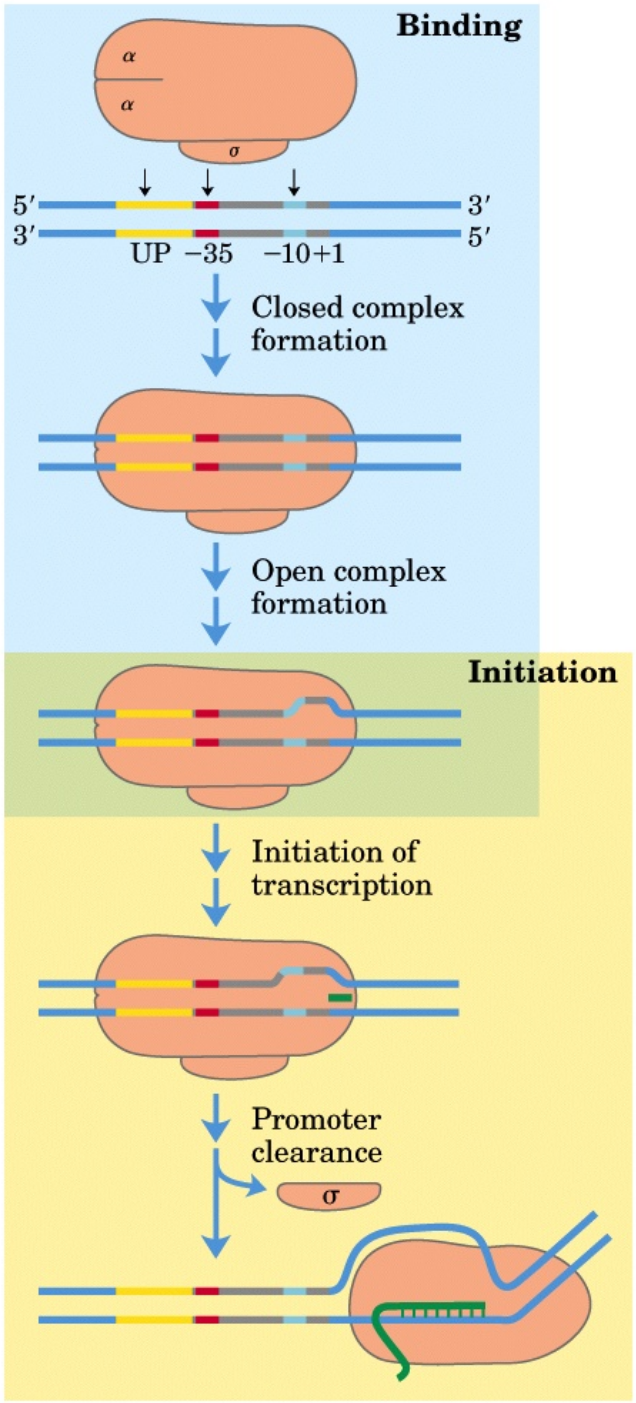
Elongation



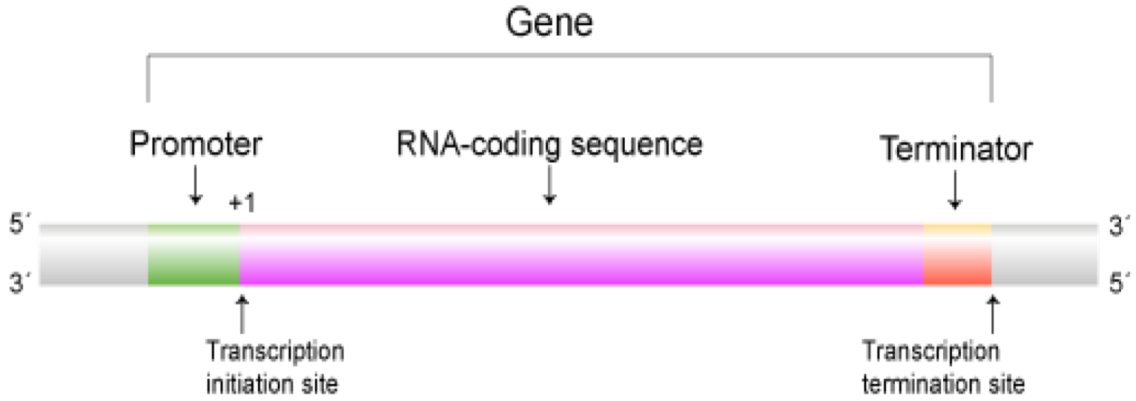
Termination



Initiation



Promoter recognition

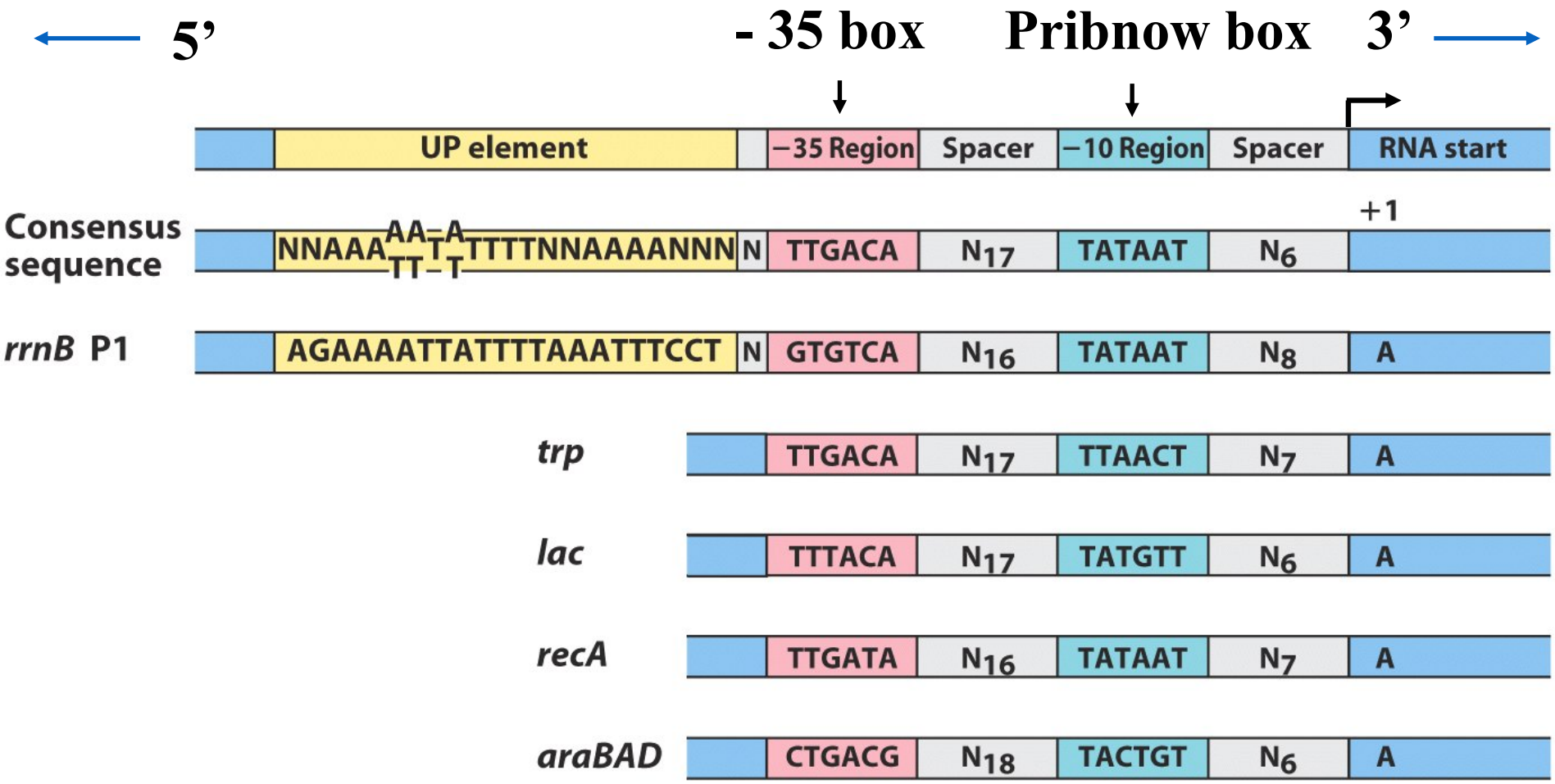


Upstream or 5’ region

Downstream or 3’ region

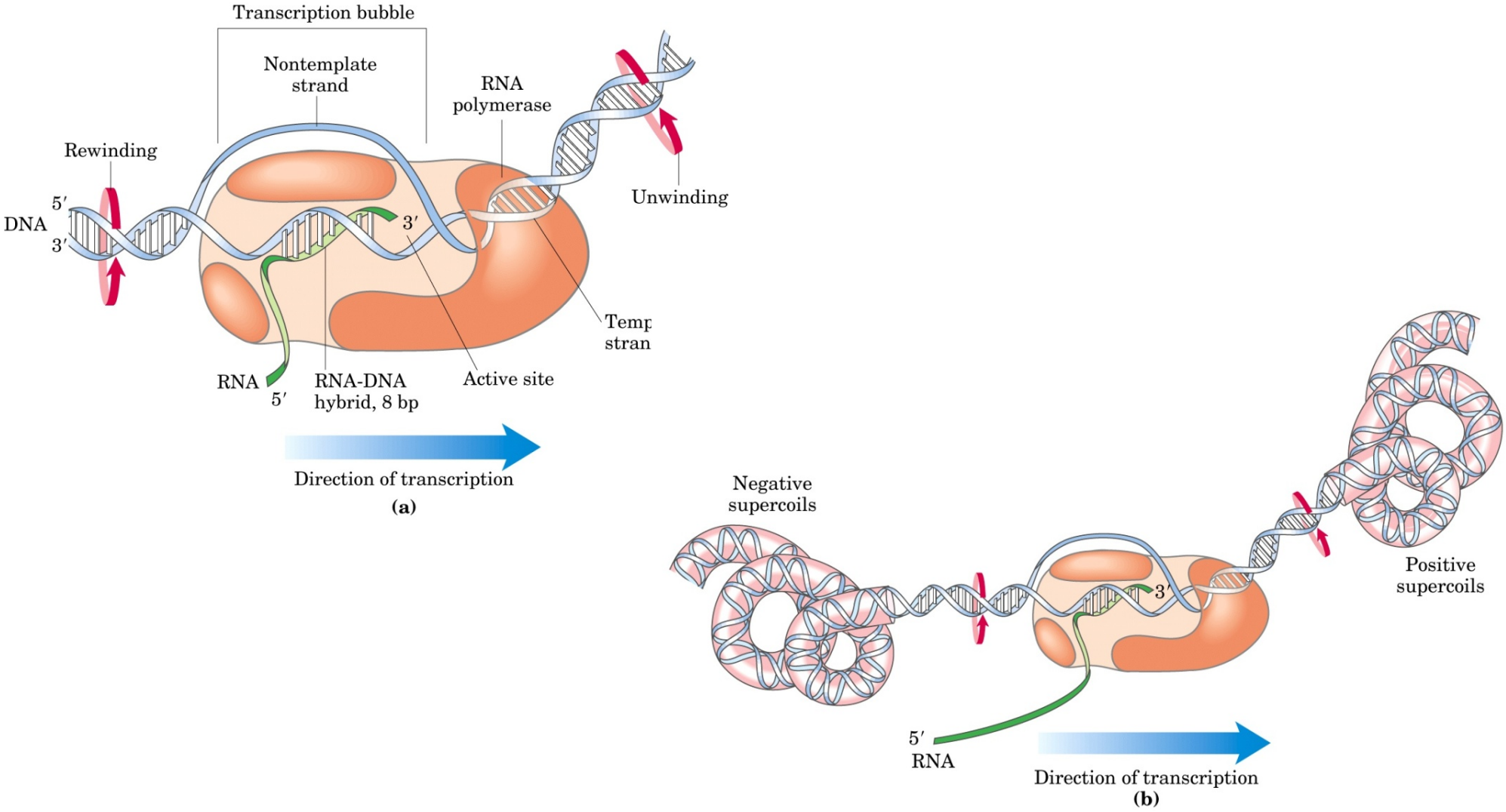
(5′) CGCTATAGCGTTT(3′)	DNA nontemplate (coding) strand
(3′) GCGATATCGCAAA(5′)	DNA template strand
(5′) CGCUAUAGCGUUU(3′)	RNA transcript

Promoter recognition

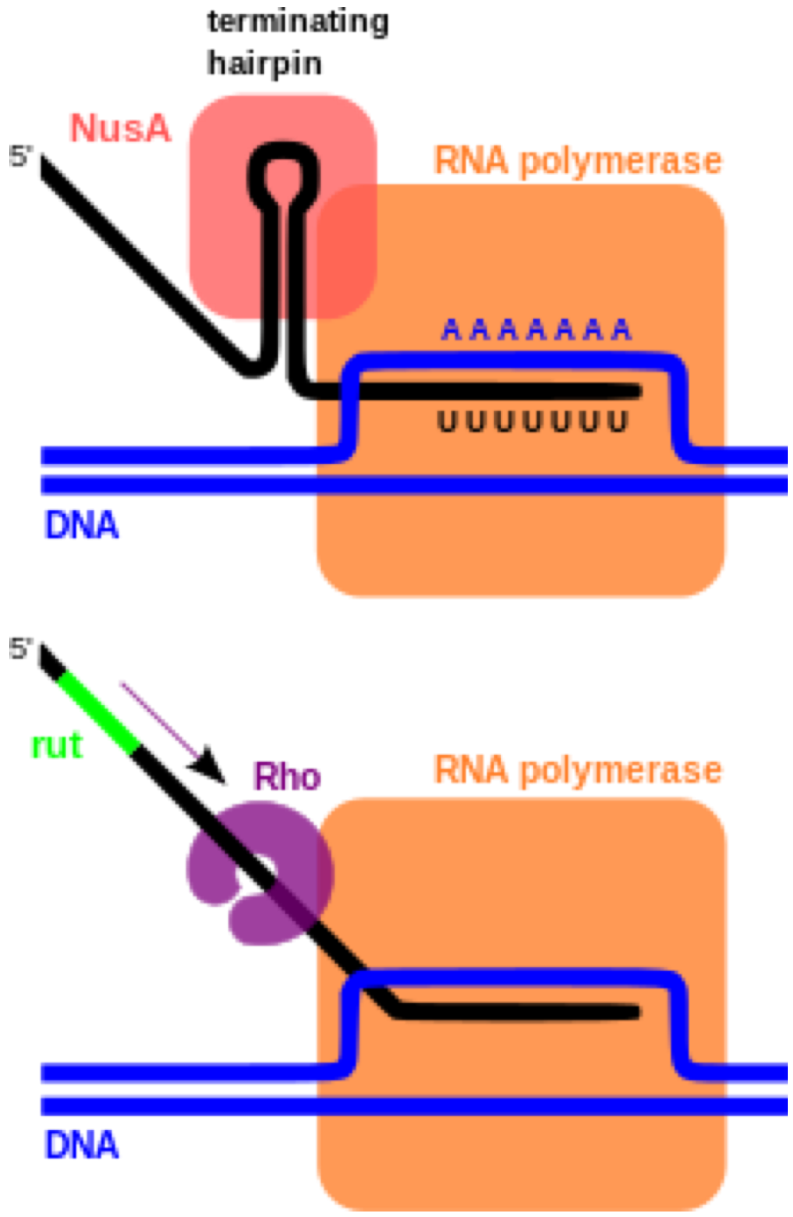


Elongation

The σ subunit dissociates from the RNA polymerase complex which leaves the promoter and proceeeds with transcription.



Termination



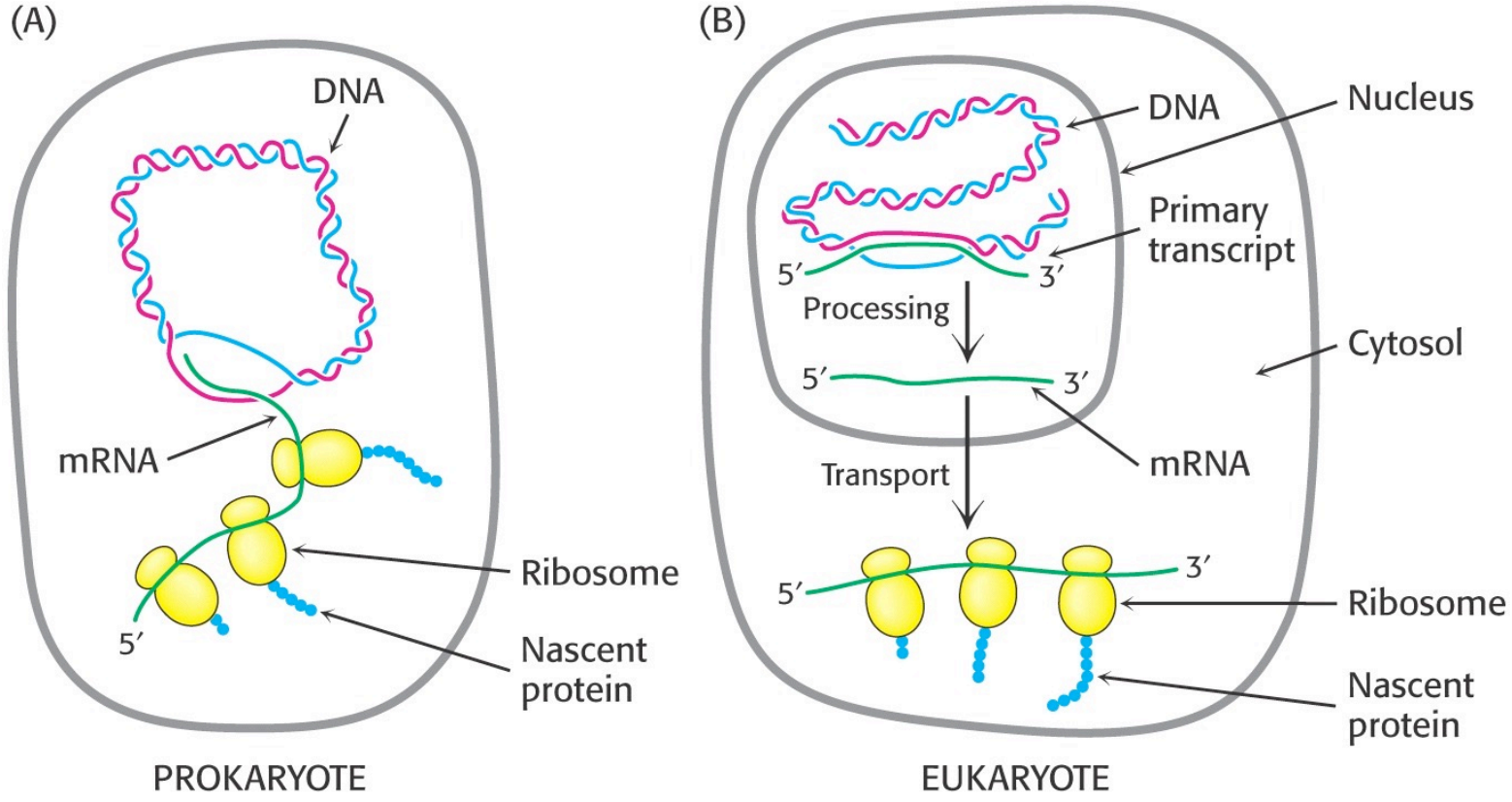
ρ–independent mechanism

ρ–dependent mechanism

Transcription in eukaryotes

Fundamental differences with prokaryotes.

1. In eukaryotes transcription and translation take place in separated subcellular compartments.



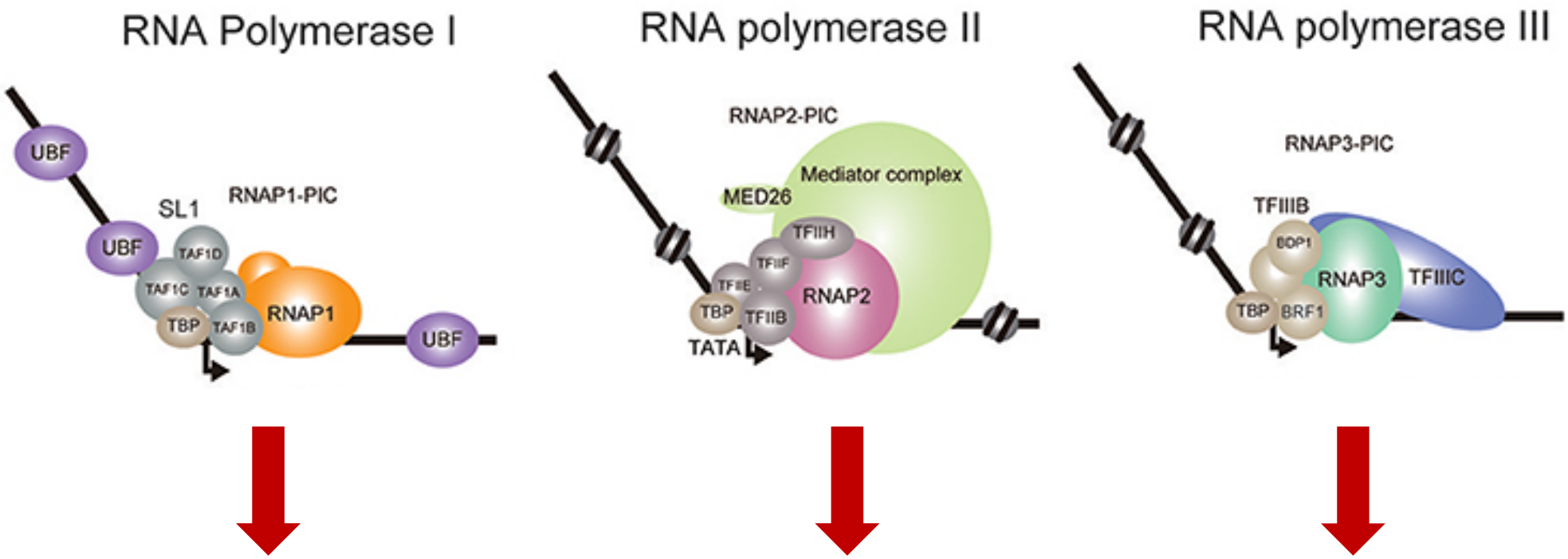
2. In eukaryotes DNA is tightly associated with proteins forming a complex tertiary structure: **chromatin**.

3. In **prokaryotes** there is **no post-transcriptional processing** of mRNA, while tRNAs and rRNAs are processed in both types of cells.

4. In **eukaryotes** nuclear genes are transcribed by **three different RNA polymerases**. There are additional RNA polymerases in mitochondria and chloroplasts of plant cells.

5. Eukaryotic polymerases do not come in direct contact with DNA, they need the participation of transcription factors.

The three eukaryotic RNA polymerases



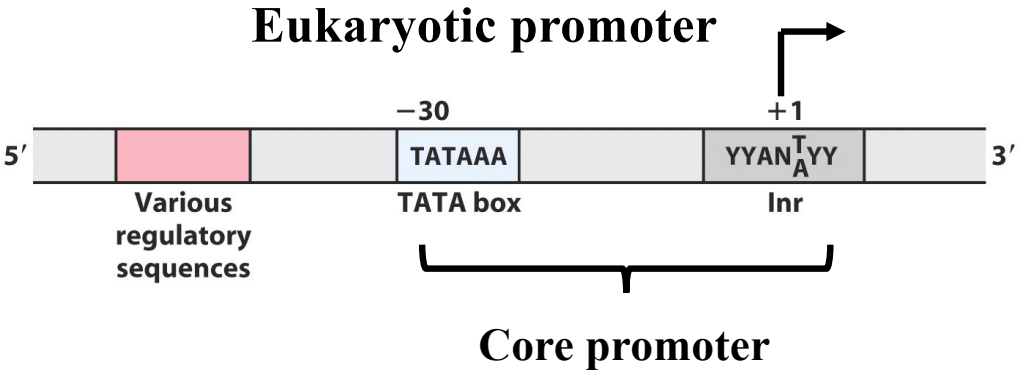
CLASS I GENES
Large pre-rRNA
28S, 18S and 5.8S
rRNAs

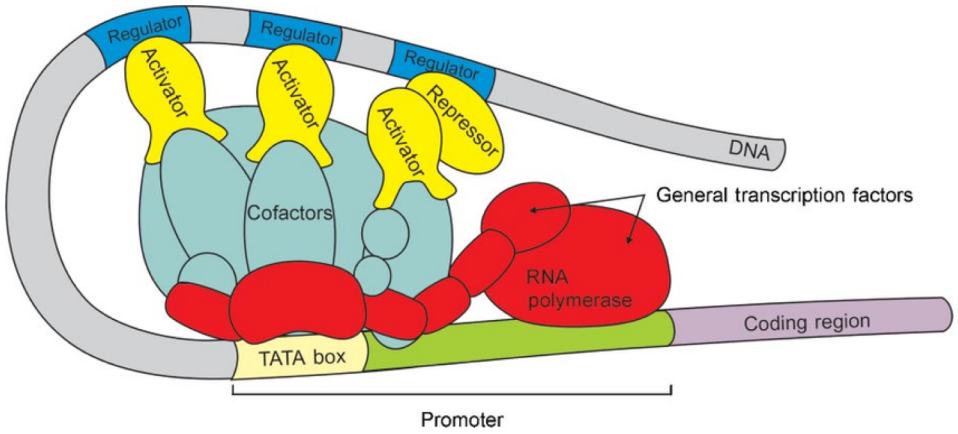
CLASS II GENES
mRNAs
and most snRNAs

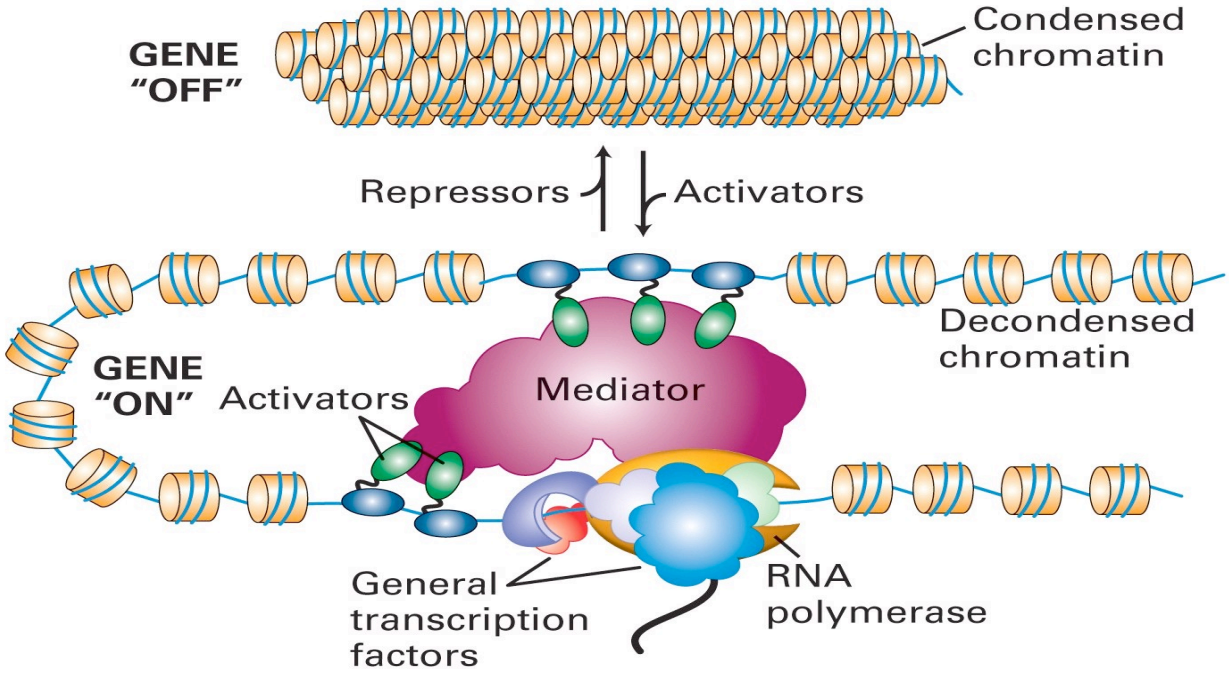
CLASS III GENES
tRNAs
5S rRNA

RNA Pol II

It requires general transcription factors (TFs) that are common to all genes. There are also gene-specific transcription factors: activators (or trans-activators) and coactivators.

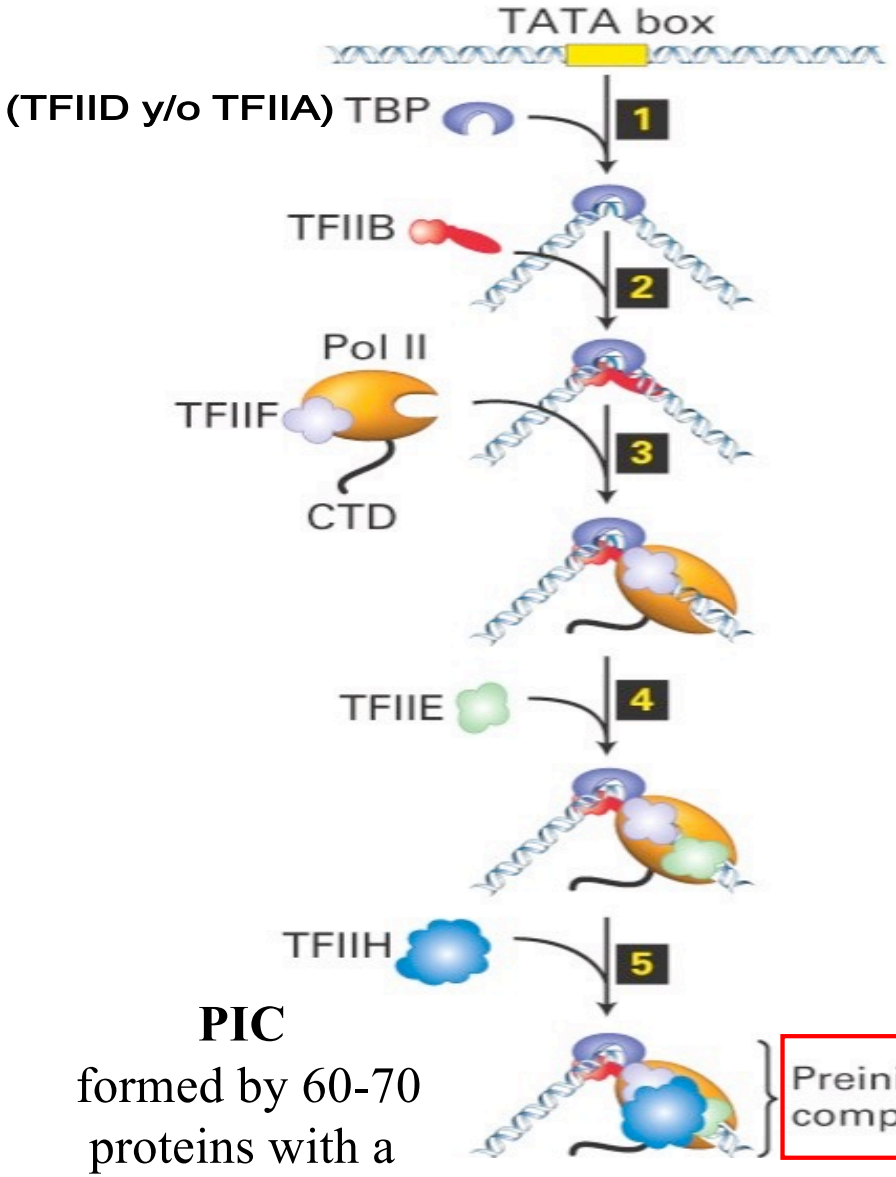






Transcription initation

Sequential binding of GTFs and RNA Pol II to an eukaryotic promoter.



PIC
formed by 60-70 proteins with a total mass ~3MDa

TFIID

Interacts with positive and negative regulatory proteins.

TFIIB

Binds to TBP and recruits the RNA Pol-TFIIF complex.

TFIIF

Binds to RNA Pol and TFIIB and prevents binding of RNA Pol to non-specific DNA sequences.

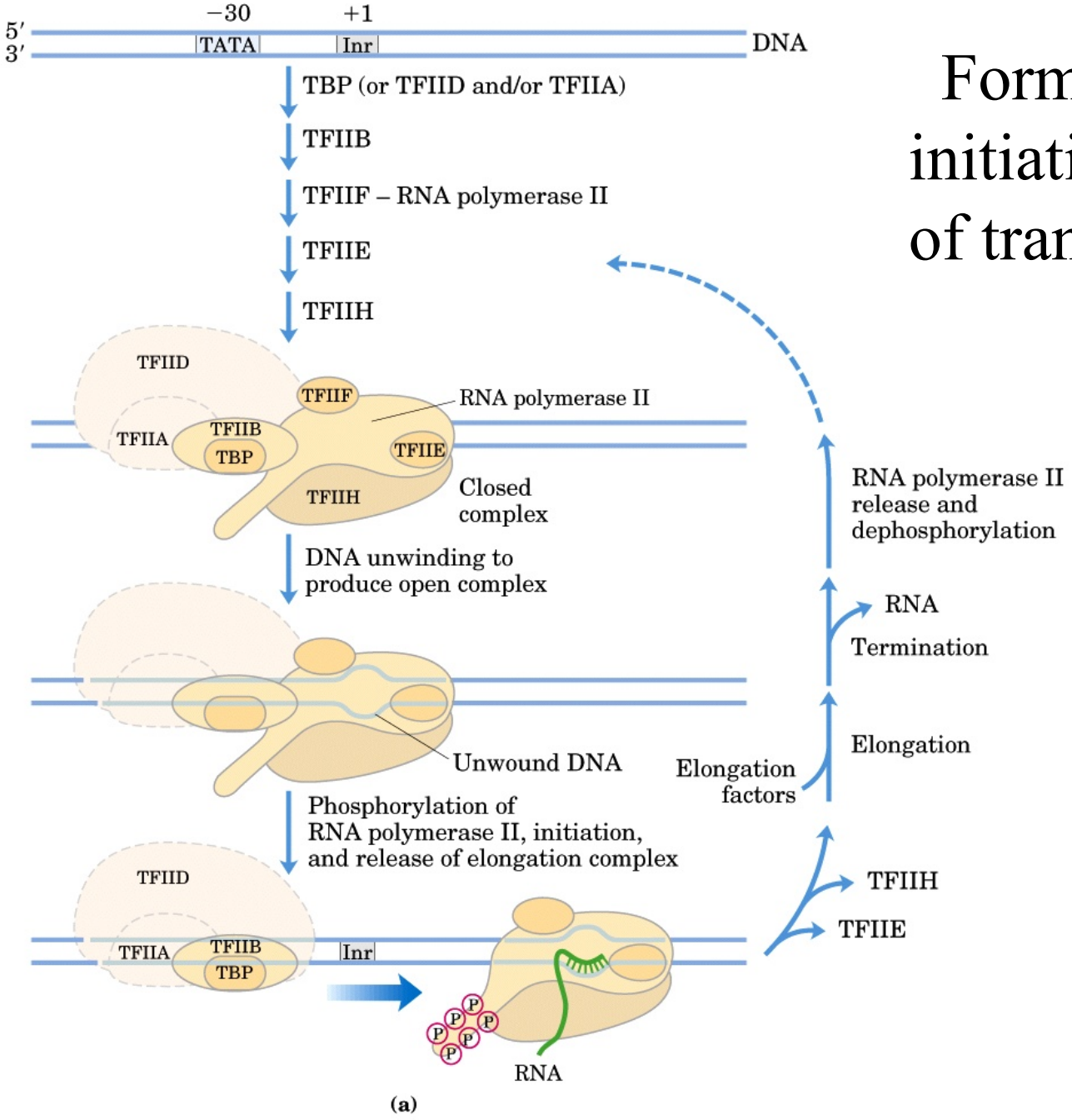
TFIIE

Helicase activity

TFIIH

Helicase activity
Kinase activity on RNA Pol CTD

Formation of the PIC, initiation and elongation of transcription by **RNA Pol II**

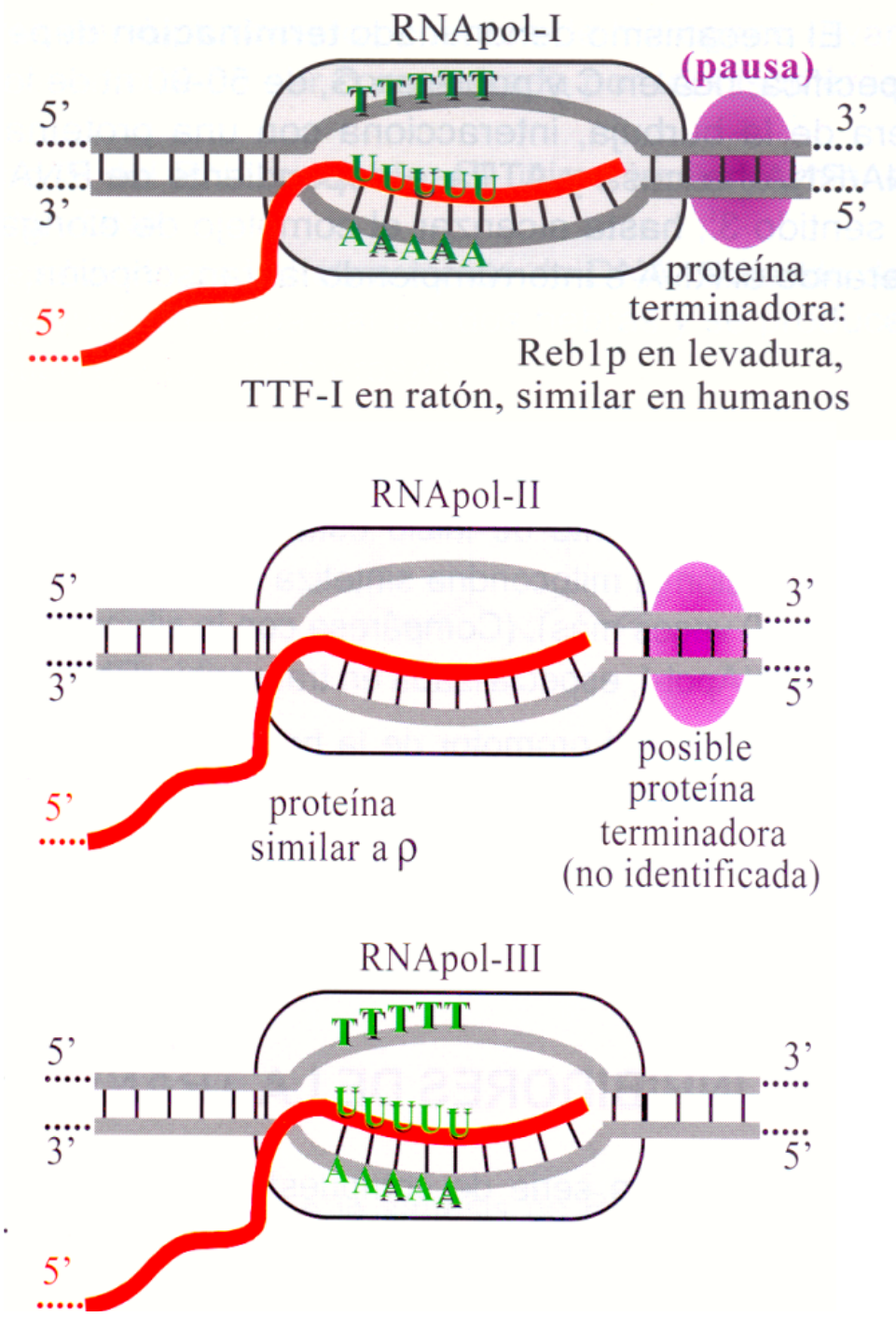


Transcription termination in eukaryotes

RNA Pol I

RNA Pol II

RNA Pol III



RNA processing

In both types of organisms, eukaryotes and prokaryotes, RNAs must be chemically modified to attain their mature and functional forms

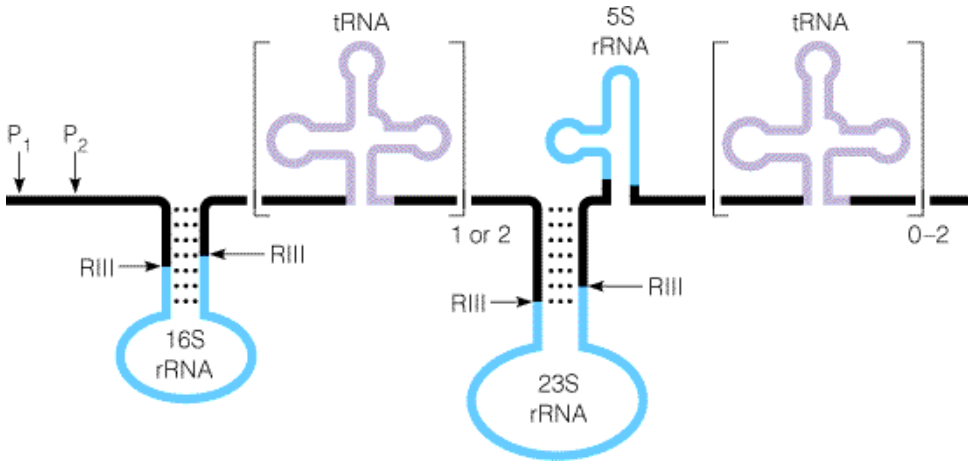
The reactions involved in RNA processing include:

- Nucleotide elimination.

- Modification of nucleobases.

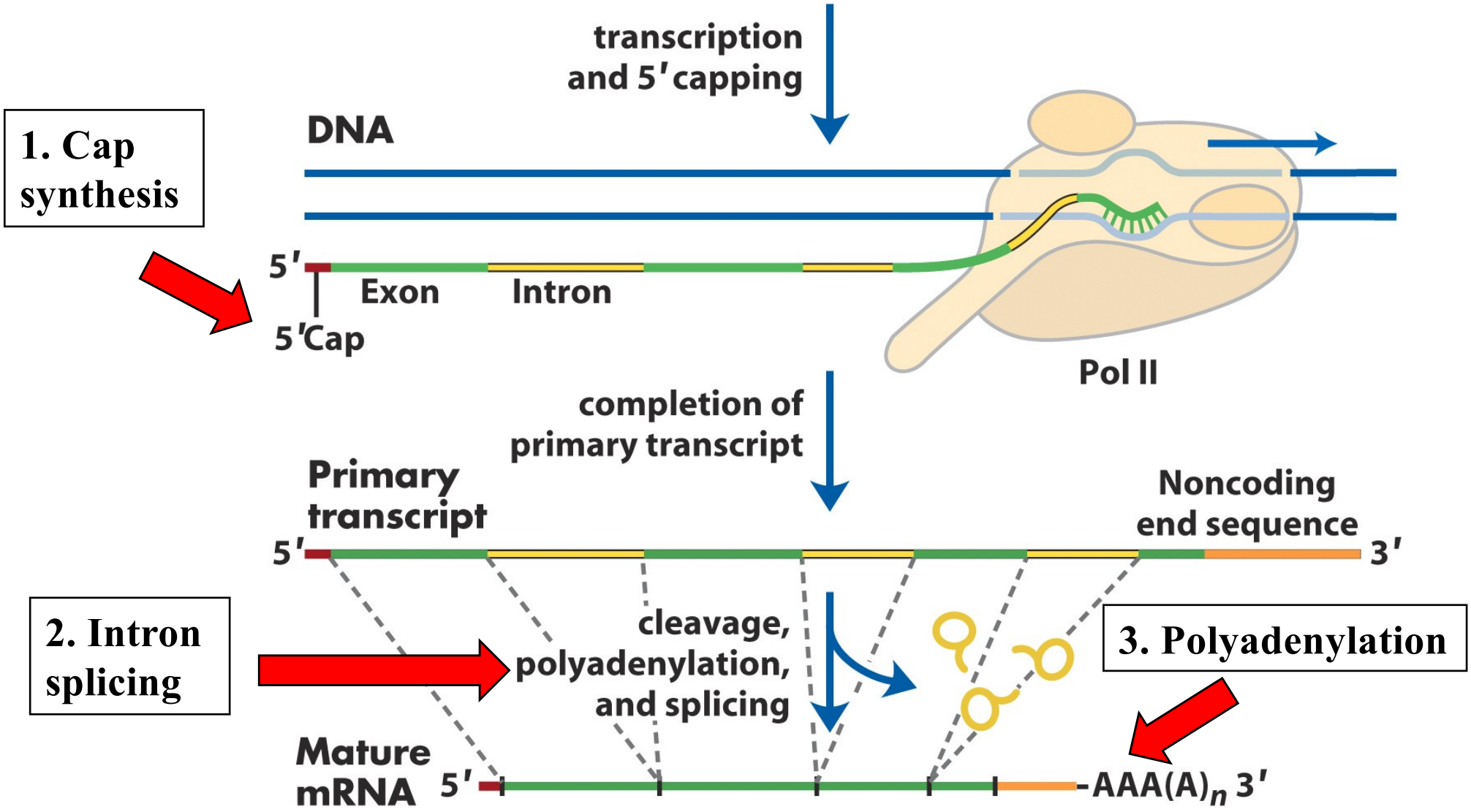
- Nucleotide addition

- Splicing reactions.

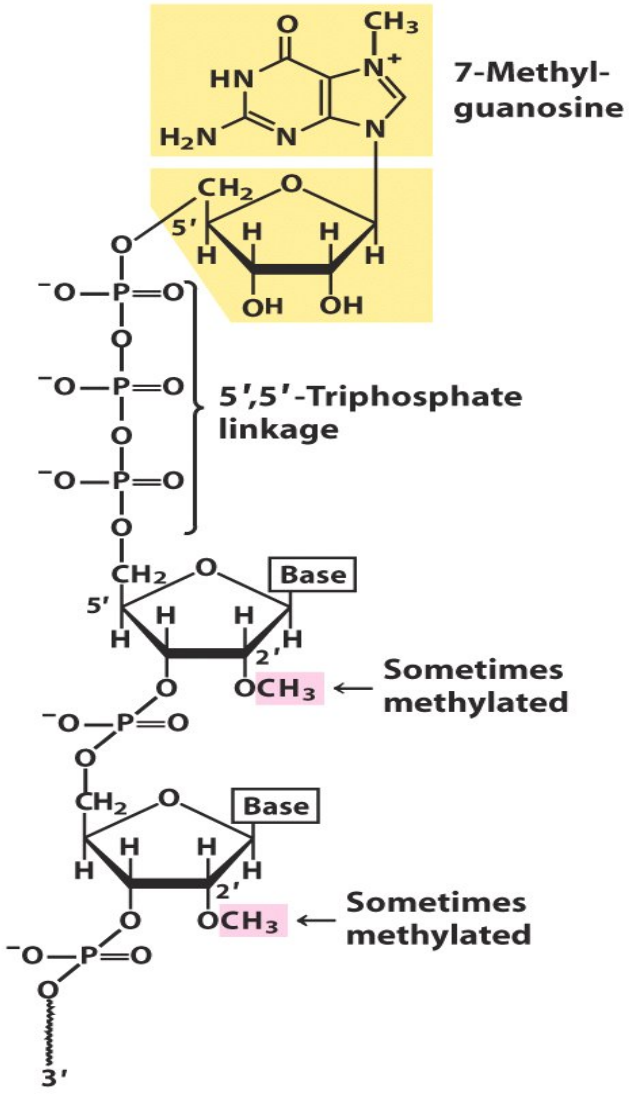


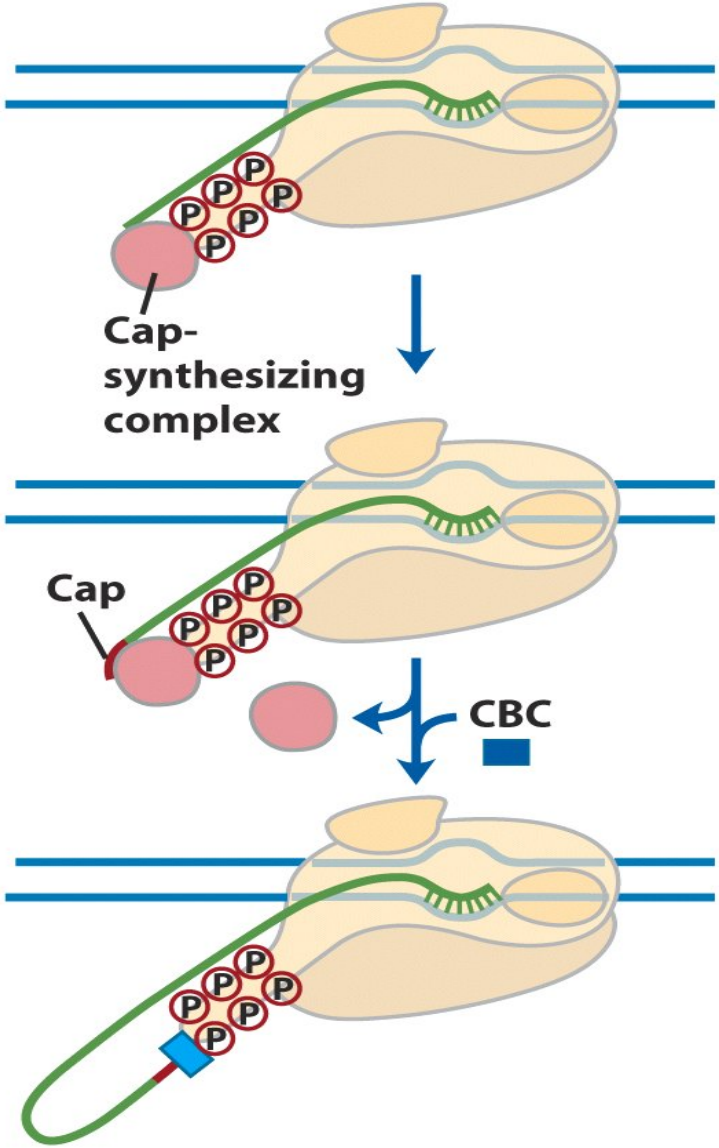
mRNA processing in eukaryotes

About ¾ of mRNA primary transcripts are eliminated during maturation in the nucleus



mRNA “capping”

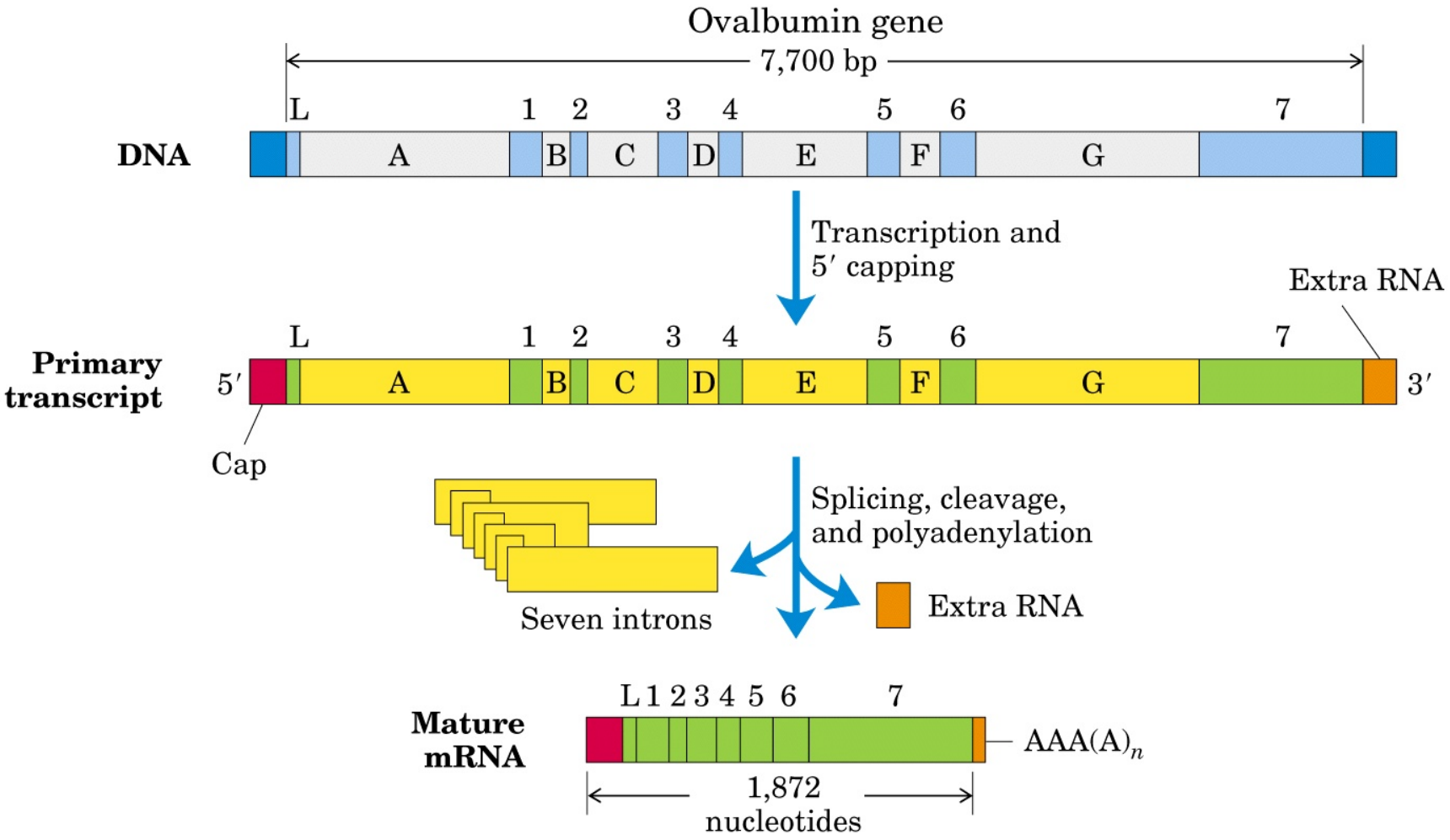




- **Protects** mRNA from degradation by exonucleases acting on the 5’ end.
- Participates in mRNA **transport** from the nucleus to the cytoplasm.
- Anchors the mRNA to the **ribosome**.

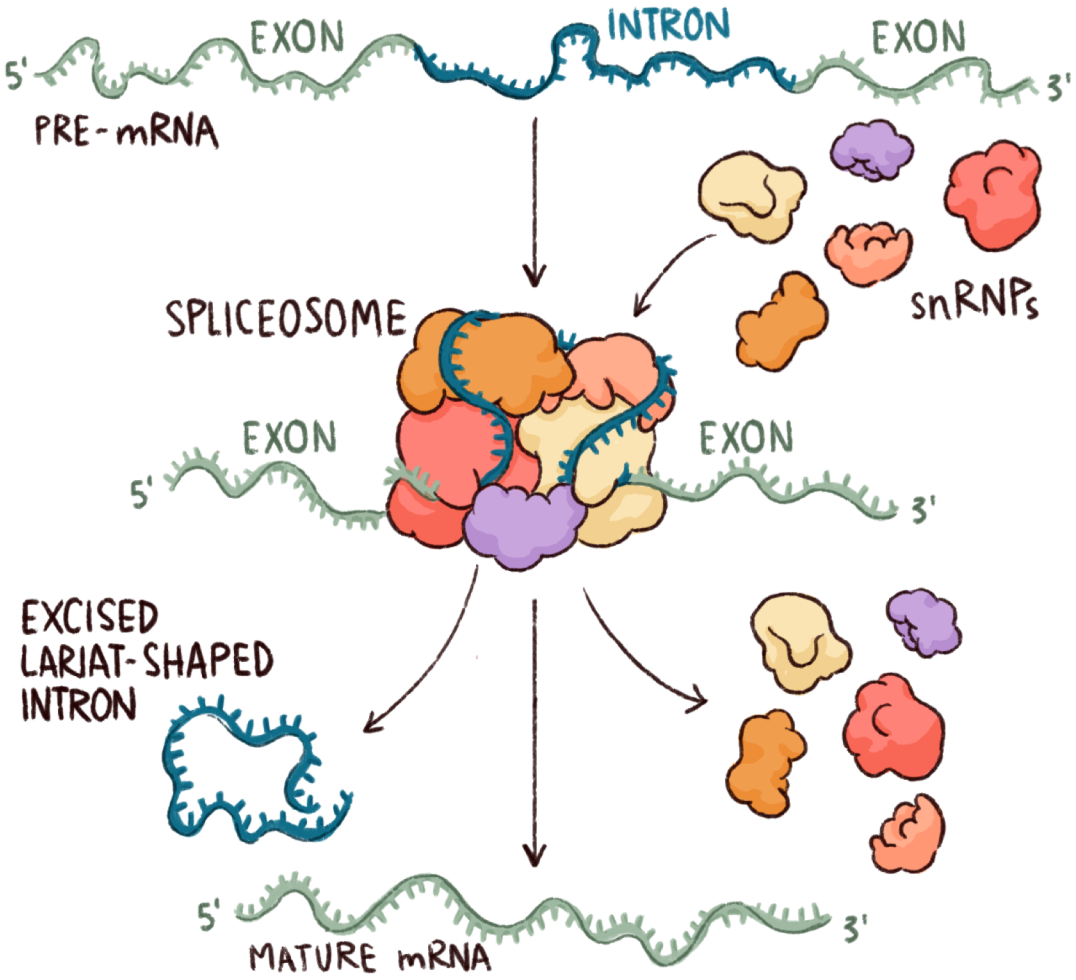
Intron elimination or mRNA “splicing”

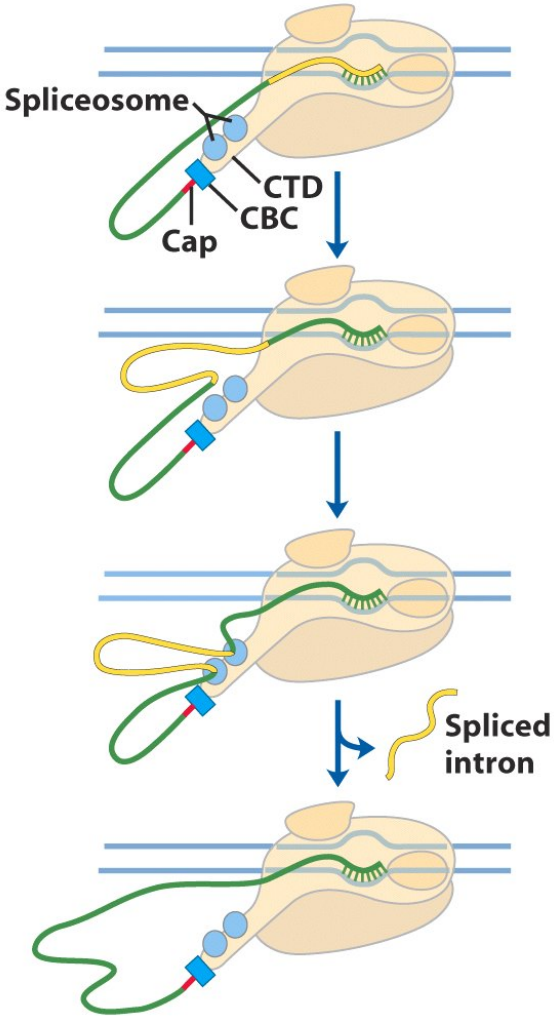
- 90% of pre-mRNA corresponds to introns.
- Average exon size: 100-200 nts, introns may be up to 20 kb.



Pre-mRNA splicing is mediated by the SPLICEOSOME

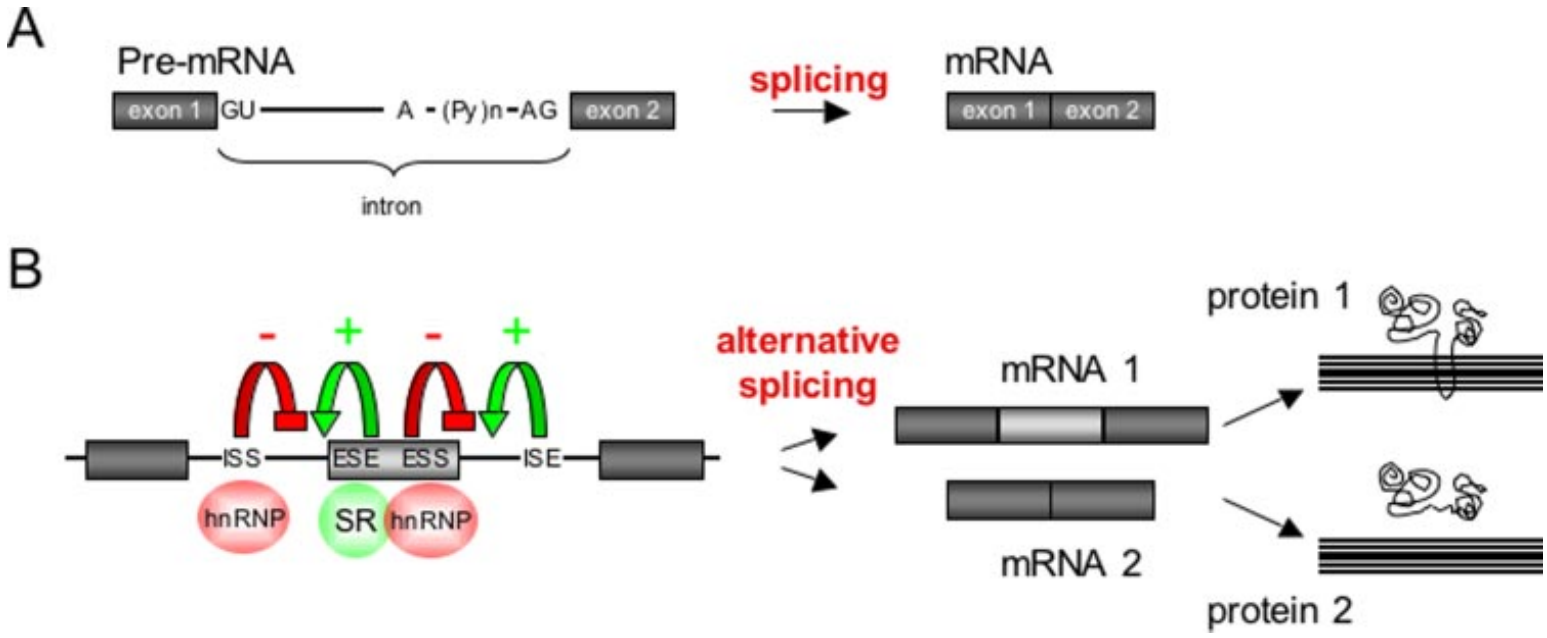
Small nuclear ribonucleoproteins **snRNPs**
~ 150 proteins and 5 small nuclear RNAs (snRNAs)





Alternative Pre-mRNA splicing

- Alternative patterns of intron removal allow the synthesis of multiple mRNAs from a single gene encoding different proteins with potentially different functions.
- This mechanism not only expands the information contained in the genome (i.e. protein variants), it also implies the existence of additional regulatory possibilities.
- Alternative splicing affects most human genes.
- Mutations and alterations in the expression of splicing regulatory factors can participate in disease development.



mRNA polyadenylation

- All mRNAs in animal cells, except histones mRNAs, have a poly-A tail in their 3' end.
- This poly-A tail is bound by specific proteins that inhibit the action of nucleases and therefore delay mRNA degradation.

